

Introduction

PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

Day 1/5

Dr. Vittorio Del Tatto

The team

- Dr. Monika Madhavi, **Dr. Marta Monti**, **Dr. Vittorio Del Tatto**



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கொழும்புப் பல்கலைக்கழகம்
UNIVERSITY OF COLOMBO



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In this summer school:

“Molecular Dynamics is like cooking: even as a user / consumer you should know the basic ingredients and be able to distinguish good and bad food ”

Structure of the days: theory (morning) + hands-on (afternoon)

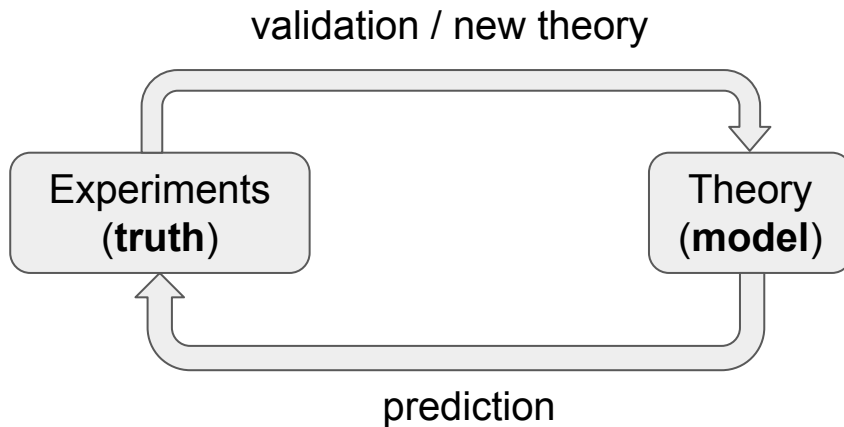
- Day 1: introduction to the field; integrating Hamilton equations, sampling statistical ensembles
- Day 2: force fields, solvent, running simulations and analyzing trajectories
- Day 3: enhanced sampling
- Day 4: ab-initio molecular dynamics
- Day 5: machine-learning integration with MD

...ask questions!!

Computational and theoretical biophysics

What do we mean by computational science?

- The scientific method in a nutshell:



- Two kinds of predictions:
 1. analytical (simple models)
 2. **numerical (complicated models)**
- Computational science **sits in the middle**: when the model is too complex to be solved analytically, numerical predictions (or **computational experiments**) are still possible
- Important disclaimer: computational experiments approximate the model, not the real world!

Biophysics in a nutshell

- Very **interdisciplinary field**; studies the behaviour and features of biological molecules using tools from Physics

Experimental

Structure:

- X-ray crystallography ($\sim 1\text{--}2\text{ \AA}$)
- CryoEM
- ...

Dynamics:

- Ultrafast pump-probe experiments ($\sim 5\text{--}10\text{ fs}$)
- Fluorescence spectroscopy
- Optical tweezers
- ...

Theoretical and computational

- Numerical methods for simulations: classical molecular dynamics (**MD**), quantum (**ab-initio MD**)
- Force fields
- Machine learning (**ML**) techniques
- Coarse-grained MD
- Active matter
- Polymer physics
- ...

- **If the model is a good approximation of reality**: simulation as *computational microscope*, time and space resolutions of the most advanced experimental techniques at a fraction of the effort

The problem of scales

- The “theory” employed depends on the **time scales** of interest and the **size** of the system

<i>Theory</i>	Quantum mechanics	Classical mechanics	
<i>Simulation methods</i>	Ab-initio MD, QM/MM	All-atom MD	Coarse-grained MD

The diagram illustrates the problem of scales in molecular simulation. It shows three representations of a water molecule (H₂O). The leftmost representation is a quantum mechanical model (Ab-initio MD, QM/MM) showing electron density isosurfaces in blue and green around the oxygen (O) and hydrogen (H) atoms. The middle representation is an all-atom molecular dynamics model (All-atom MD) showing the water molecule as a central red sphere (O) bonded to two white spheres (H). The rightmost representation is a coarse-grained molecular dynamics model (Coarse-grained MD) showing the water molecule as a single purple sphere (O) with two smaller purple spheres (H) attached, all enclosed within a dashed purple boundary representing the coarse-grained bead. Arrows point from the 'Ab-initio MD, QM/MM' and 'All-atom MD' simulation methods to the quantum and all-atom models, respectively. An arrow points from the 'Coarse-grained MD' simulation method to the coarse-grained model.

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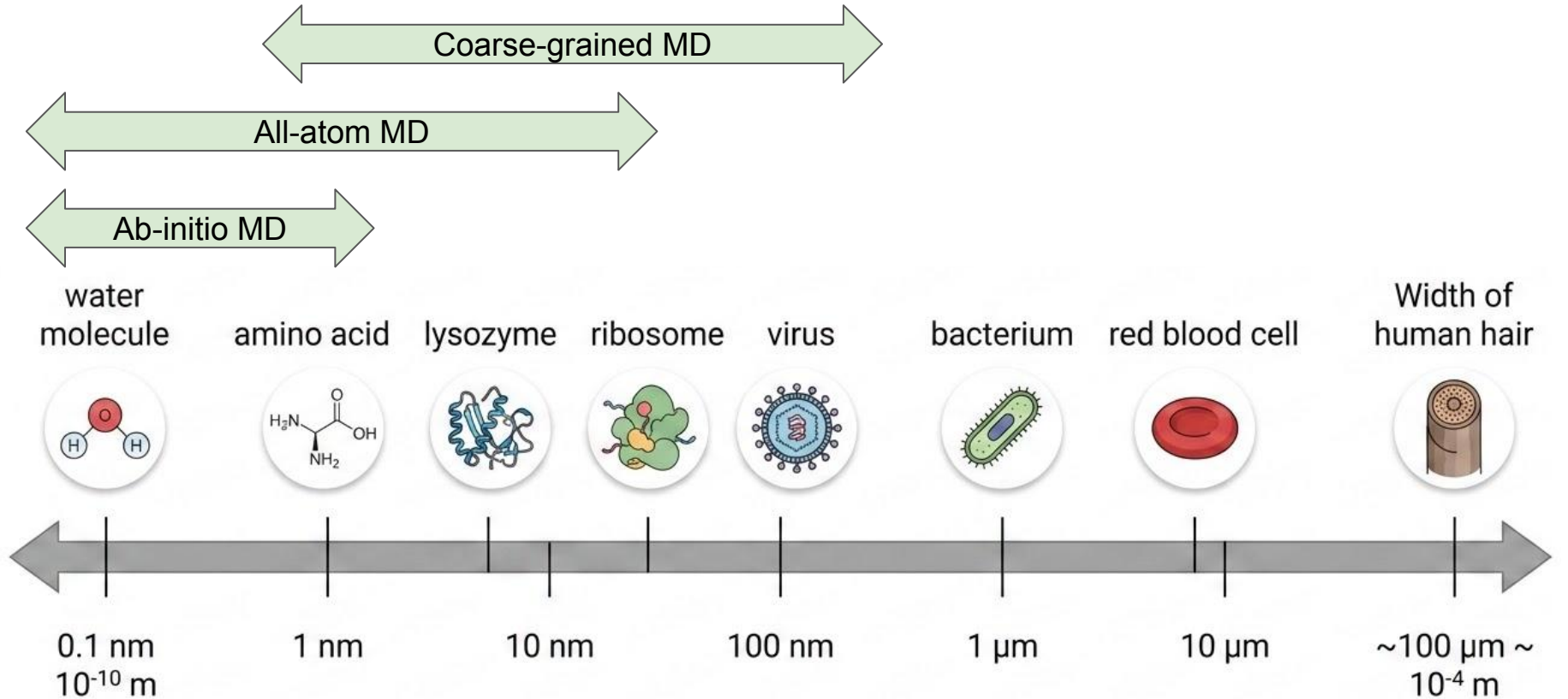
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<i>Processes resolved</i>	Covalent bond breaking, electron transfers, transitions to excited states	Atomic vibrations, H-bonds formation / breaking, side-chain rotations	Soft collective modes (e.g. bending of a membrane)

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<i>Time resolution</i>	~ 0.1–0.5 fs	~ 1-2 fs	~ 20-40 fs
<i>Total time reachable (routine)</i>	~ 10–100 ps	~ 1-10 μ s	~ 10 μ s –1 ms
<i>Typical system size</i>	10^2 – 10^3 atoms	10^4 – 10^6 atoms	10^5 – 10^7 beads (10^6 – 10^8 atoms)

Length scales in biology



Classical vs ab-initio MD – key differences

	Classical MD	Ab-initio MD
<i>Law of motion</i>	Newton's equation	Schrodinger's equation for e^- ; Newton's equation for nuclei

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<i>Numerical integration</i>	Verlet integrator	Self-consistent energy calculation for electrons, Verlet integrator for nuclei

Statistical ensemble

- Why not only Newton's equations? Because they describe a system which is:
 - **closed** (no exchange of matter with environment)
 - **isolated** (no exchange of energy)
- } $\longleftrightarrow$ total energy conserved



Almost never the case in a real experiment!

- More realistic systems (not isolated):
 - **closed but with exchange of thermal energy**
 - **closed but with exchange of thermal and mechanical energy**
- We'll see that specific algorithm can be used to “extend” Newton's equations and satisfy the scenarios above: **thermostats** and **barostats**

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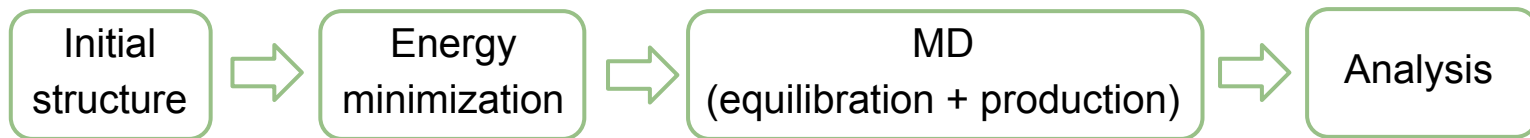
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Other ingredients

- Simulation parameters: **time step** of integrator, trajectory length (simulated time), ...
- Approximations for **long-range interactions**
- How to treat **hard** and **soft constraints** (e.g. rigid bonds, fixed distances, or stiff springs)
-

Full simulation pipeline:



- Today several simulation softwares are available

MD softwares

Classical MD engines:

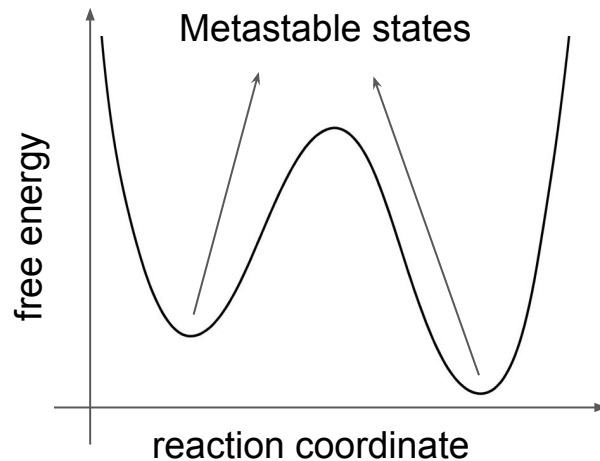
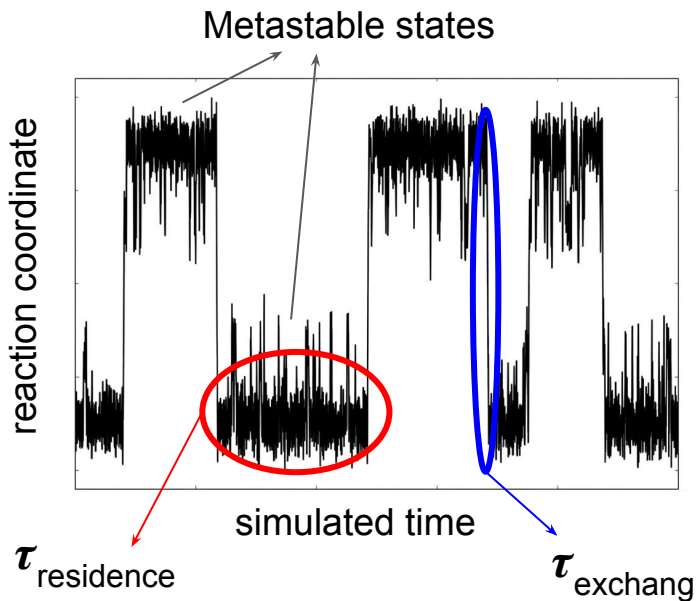
- Gromacs (what we'll use in these days) – more liked by biophysics community
- LAMMPS – more liked by solid state physics community
- ...

Ab-initio MD:

- CP2K (what we'll use in day 4)
- Quantum Espresso
- ...

Metastability and rare events

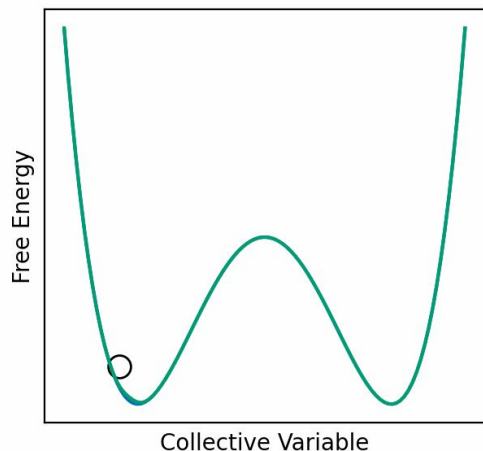
- Typical trajectory showing **metastability** and **rare transitions**:



- Ratio of these two time scales quantifies the degree of metastability
- Rare transitions described as **jumps of free energy barriers**

Enhanced sampling

- Problem of rare events: require long trajectories to be observed (often too long than what achievable with all-atoms MD)
- Problem tackled by **enhanced sampling** methods
 - make rare events *frequent in the simulation*
 - e.g. umbrella sampling, metadynamics, replica-exchange, transition path sampling, ...



Taken from: B. Hishberg, "Stochastic Resetting for Enhanced Sampling"
<https://hirshlab.sites.tau.ac.il/post/stochastic-resetting-for-enhanced-sampling>

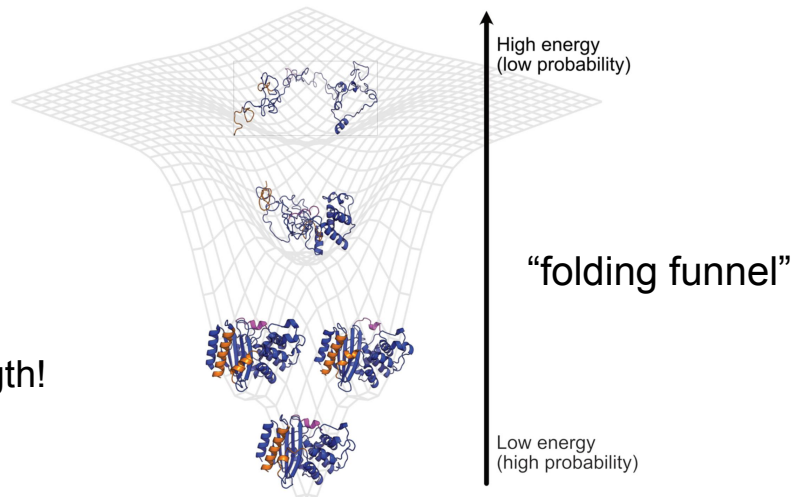
A challenging example of metastability: protein folding

Example: protein folding

- $\tau_{\text{exchange}} \equiv \text{folding} \sim 10\text{-}100 \text{ ns}$
- $\tau_{\text{residence}} \sim 1 \text{ ms}$



only few supercomputers / computational frameworks can achieve in practice this length!



Taken from: Bowman GR, Annu Rev Biomed Data Sci. (2024)

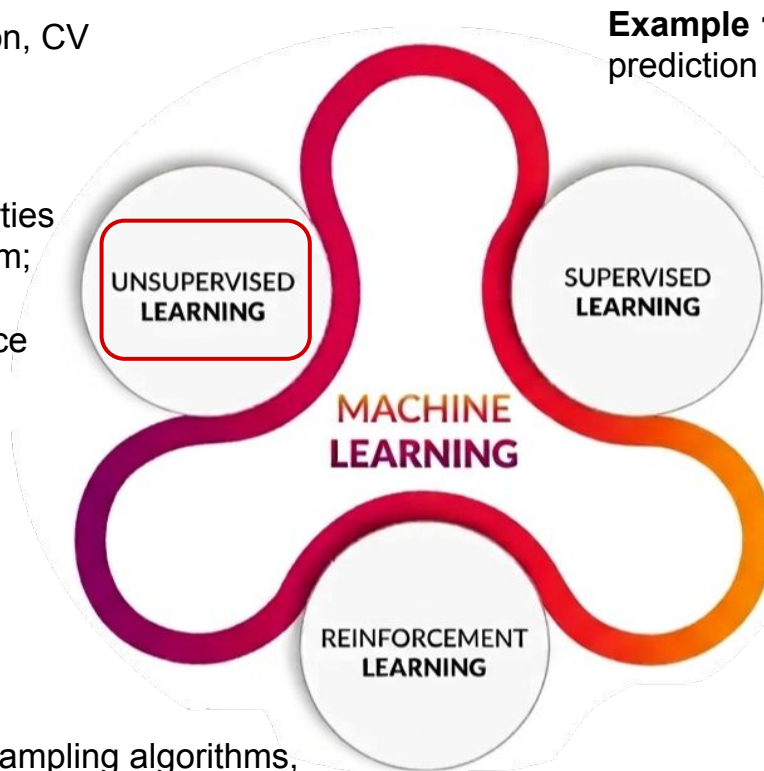
- N.B. **AlphaFold** (2024 Nobel Prize in Chemistry) “solved” the problem of protein structure prediction, not the full protein folding problem (no kinetic information!)
- **Why** the problem is still **unsolved**?
 - Enhanced sampling usually requires good reaction coordinates, unknown a priori
 - Errors in the force field can significantly increase the escape time (height of barriers)

ML in molecular simulations

Examples: data visualization, CV construction, ...

↓
Goal: understanding properties of a high-dimensional system; learning variables that can describe it in a low-dim space

Examples: sampling algorithms, molecule design



Example 1: ML force-fields; graph NNs for prediction of electronic properties

↓
Goal: substituting Ab-initio MD / expensive QM calculations with cheaper tools, preserving the accuracy

Example 2: Selecting a subset of features in a candidate high-dimensional set of molecular descriptors

↓
Goal: reducing the dimensionality of the representation to make plots or use other analysis methods